



A Look at OpenSSH, Netcat and PF

This article deals with some of the systems administrator's most familiar tools.

OpenSSH was a project started by the OpenBSD team based on the work done by Tatu Ylonen, a Finn, who wrote something called SSH (Secure Shell) back in 1995. It came as a welcome replacement to telnet, which has been a security nightmare for several decades. Today, SSH and its companion, SCP (Secure Copy), can work wonders, not only from a security standpoint, but also from a convenience angle. Plenty of new features are being added with each release to enhance the security with strong crypto, more user conveniences like printing the fingerprint graphics, and so on. But for the purpose of this article, I will focus only on certain specific use cases. I have found that SSH is an incredibly powerful, versatile and capable tool that lends itself easily to automation and quickly getting a job done. You can even set up remote or local port forwarding, which is normally only done by firewalls.

Let's look at some of the huge variety of things SSH can do. At the simplest level, it is used for logging into a (usually remote) system with `ssh user@host`, after which you're prompted for a password and are logged in. If you set up public key authentication (described below) and load

your private key using the `ssh-agent` daemon, then you can log in without any password—very convenient for non-interactive scripts that run remotely. This works well for SCP too, which uses the same network protocol as SSH, but does file transfer instead. Another program, SFTP, uses FTP-like semantics, but differs since it uses secure connections, has no multiple connections like data and command connections, or active and passive modes like conventional FTP. I mostly use only SSH or SCP.

Running a simple remote command using SSH is easy: `ssh foo.bar ls`, for example, which connects to the `foo.bar` machine and lists the files in the default folder. If you don't supply a username with an `@` before the hostname (or with the `-l` switch), then SSH uses the username under which you are logged in to the local machine. The default folder is then the home folder for that user account on the remote host.

However, for editing and other interactive commands, this approach will not work. For example, you have to run SSH as follows: `ssh -t girish@yahoo.com vi /etc/ntp.conf`, in which `-t` switch sets up the terminal mode necessary for working transparently on the network. Once you exit `vi`, you are back at the local machine prompt, since the SSH session is torn down on completing the editing command.